H&M Retail Intelligence

Exploratory Data Analysis (EDA) report: graphs, descriptions and benefits

Source: *sic_finalproject_FINAL.ipynb* (SIC 8, AI Track) | Dataset: H&M Personalized Fashion Recommendations (Kaggle)

Please read: what is and is not in this report

The notebook has EDA in three places. Only **Part 1** was run with its results saved, so the **7 graphs in this report are all the graphs that exist in the notebook file**. They are shown here exactly as saved, and every number comes from the notebook's own printed outputs.

Part 2 (Extended EDA) and the EDA half of **Part 3 (Churn features)** contain the code for about 30 more graphs, but their cells have **no saved output** (not run, or outputs cleared). I did not invent results for them. Section 3 lists each of those graphs with what it will show and why it is useful; to get the actual pictures, run the notebook on Kaggle (Save & Run All) and send me the executed .ipynb.

Section	Content	Graphs in file
1	Dataset overview and catalogue distributions (tables)	none (printed tables)
2	Part 1 EDA with graphs: time trends, missing values, correlation, skewness	7
3	Part 2 and Part 3 EDA: planned graphs, no saved output	0 saved (about 30 coded)
4	Key takeaways from the executed EDA	-

1. Dataset overview

The full Kaggle data was explored first (Part 1). All numbers below are from the full data unless marked as sample.

Table	Rows x columns	Notes
transactions	31,788,324 x 5	t_dat, customer_id, article_id, price, sales_channel_id. No missing values. 1,362,281 distinct customers and 104,547 distinct articles appear in it.
customers	1,371,980 x 7	customer_id, FN, Active, club_member_status, fashion_news_frequency, age, postal_code.
articles	105,542 x 25	Product catalogue: type, group, colour, pattern, department, section, description.

- **Date range:** 20 Sep 2018 to 22 Sep 2020. There are 734 distinct dates, which equals the number of calendar days in that range, so the daily series has no missing days.
- **Sales channel:** channel 2 = 22,379,862 transactions (70.4%), channel 1 = 9,408,462 (29.6%).
- **Price:** all prices are above 0. Mean 0.0278, median 0.0254, max 0.5915 (prices are scaled, not real currency).

Benefit. Knowing sizes, types and date range early shows the data is too big to model directly (31.8M rows), so a 10% customer sample was used for the rest of the project; it confirms the time axis is complete, which is required for a time-based train/test split.

1.1 Catalogue distributions (articles table)

These come from `value_counts()` in cells 16 to 22. They count **articles in the catalogue**, not sales.

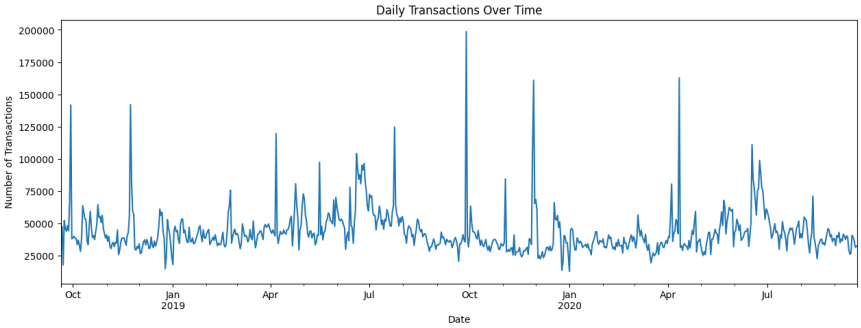
Feature	Top values (number of articles)	Reading
product_group_name (19 values)	Upper body 42,741 | Lower body 19,812 | Full body 13,292 | Accessories 11,158 | Underwear 5,490	Upper-body garments are 40.5% of the catalogue; 10 groups have fewer than 130 articles.
product_type_name (131)	Trousers 11,169 | Dress 10,362 | Sweater 9,302 | T-shirt 7,904 | Top 4,155	A handful of types dominate.
colour_group_name (50)	Black 22,670 | Dark Blue 12,171 | White 9,542 | Light Pink 5,811 | Grey 4,487	Black alone is 21.5%; dark/neutral colours lead.
index_group_name (5)	Ladieswear 39,737 | Baby/Children 34,711 | Divided 15,149 | Menswear 12,553 | Sport 3,392	Ladieswear and Baby/Children are 70.5% of the catalogue.
graphical_appearance_name (30)	Solid 49,747 | All over pattern 17,165 | Melange 5,938 | Stripe 4,990 | Denim 4,842	Plain items are 47.1%.

Benefit. Shows which categories are heavily imbalanced. Rare categories (for example 'Fun' with 2 articles or 'Furniture' with 13) are too small to learn from and should be grouped as 'Other'; dominant attributes (black, solid, upper body) will dominate any model that uses them unless they are handled.

2. Part 1 EDA with graphs (7 graphs)

Graph 1: Daily Transactions Over Time

Notebook cell 26



What it shows. Number of transactions for each of the 734 days (line chart).

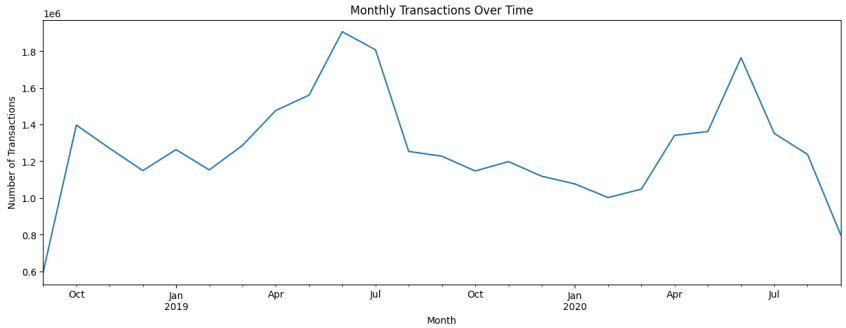
What we see in it.

- A normal day has roughly 30,000 to 50,000 transactions (median 39,515, mean 43,308, minimum 12,760).
- There are sharp one-day spikes. The highest is 198,622 transactions (read from the graph: late Sep 2019). Other spikes of about 140,000 to 165,000 appear in Sep 2018, Nov 2018, Nov 2019 and Apr 2020.
- Activity is higher around late spring and summer (May to Jul) in both years. Smaller day-to-day swings are visible between the spikes.
- *Interpretation (not confirmed by the data):* the one-day spikes look like sale or promotion days.

Benefit. Reveals extreme days that would distort averages and popularity counts. It supports using a time-based split (never random) and recency features, and it flags the spike days as something to inspect or down-weight before modelling.

Graph 2: Monthly Transactions Over Time

Notebook cell 27



What it shows. The same data aggregated by month, which removes daily noise and shows the seasonal shape.

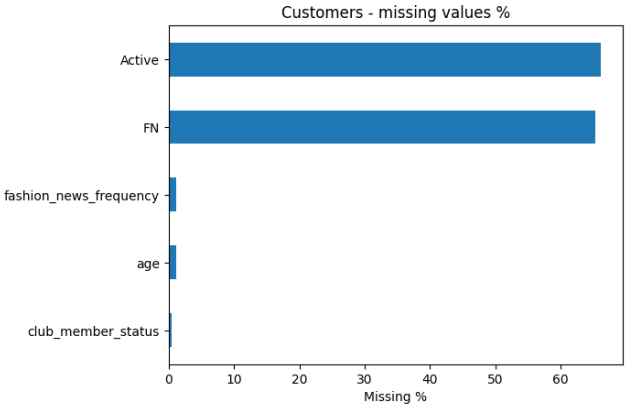
What we see in it.

- Clear yearly seasonality: highest in spring/summer (about 1.9 million around Jun 2019 and about 1.77 million around Jun 2020) and lowest in winter (about 1.0 million around Feb 2020).
- The 2020 peak is lower than the 2019 peak, and spring 2020 months (Mar to May) are also below the same months in 2019.
- The first point (Sep 2018, about 0.6 million) and the last point (Sep 2020, about 0.8 million) are partial months, because the data starts on 20 Sep and ends on 22 Sep. They are not real drops.

Benefit. Confirms seasonality, so month/season features are worth adding and recent popularity should beat all-time popularity in a recommender. It also warns us to ignore the partial first and last months when reading trends.

Graph 3: Customers: Missing Values (%)

Notebook cell 44 (1st plot)



What it shows. Percentage of missing values per column in the customers table (only columns with gaps are drawn).

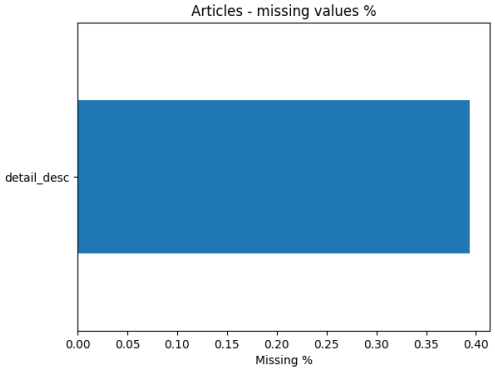
What we see in it.

- **Active** is missing for about 66% of customers and **FN** for about 65%.
- **fashion_news_frequency** (about 1.2%), **age** (about 1.2%) and **club_member_status** (about 0.4%) have only small gaps.
- FN and Active each have a single non-missing value (1.0) in the whole table, so a missing value means 'flag not set', not 'unknown'. The cross-table in cell 41 supports this: customers with FN = 1 are almost all 'Regularly' subscribed (47,314 of 47,471).

Benefit. Tells us how to treat each column. FN and Active can be filled with 0 (this is what the cleaning step does); small categorical gaps become 'Unknown'; age gaps are imputed instead of deleting customers. No rows are lost and no wrong assumptions are made.

Graph 4: Articles: Missing Values (%)

Notebook cell 44 (2nd plot)



What it shows. Percentage of missing values per column in the articles (product) table.

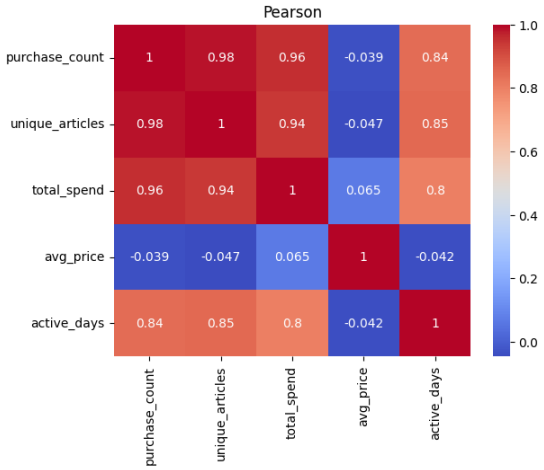
What we see in it.

- Only **detail_desc** has gaps: about 0.39% (416 of 105,542 articles).
- The transactions table has no missing values at all (the notebook prints this instead of drawing a plot).

Benefit. The catalogue and transaction data are almost complete, so all product attributes can be used as features. The small description gap is filled with 'Unknown' and has no effect on modelling.

Graph 5: Correlation Between Customer Metrics (Pearson)

Notebook cell 48 (1st plot)



What it shows. Heatmap of linear correlation between five per-customer metrics: purchase_count, unique_articles, total_spend, avg_price and active_days.

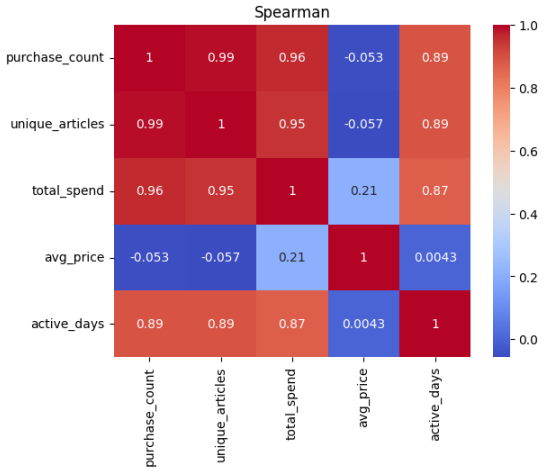
What we see in it.

- purchase_count, unique_articles, total_spend and active_days are strongly related: 0.80 to 0.98 (for example purchase_count vs unique_articles = 0.98, vs total_spend = 0.96, vs active_days = 0.84).
- avg_price is almost uncorrelated with everything (-0.05 to 0.07).

Benefit. Shows which features carry the same information. The four activity metrics are redundant, so keep one or combine them for linear models and clustering (multicollinearity). avg_price is independent and adds new information about price level, so it should be kept.

Graph 6: Correlation Between Customer Metrics (Spearman)

Notebook cell 48 (2nd plot)



What it shows. The same heatmap using rank (Spearman) correlation, which does not assume a straight-line relationship and is robust to extreme values.

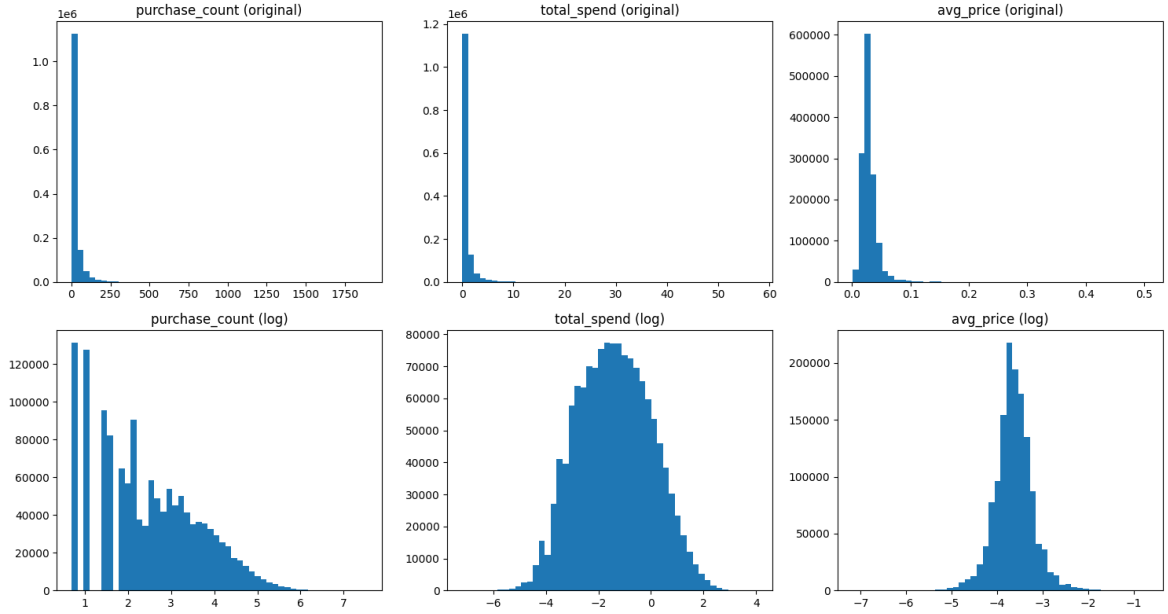
What we see in it.

- Same pattern, slightly stronger: purchase_count vs unique_articles = 0.99, vs active_days = 0.89 (Pearson 0.84).
- total_spend vs avg_price rises from 0.065 (Pearson) to 0.21 (Spearman): a real but non-linear link that Pearson hides.

Benefit. Because the data is heavily skewed (Graph 7), Spearman is the more trustworthy of the two. Comparing both confirms the same redundancy conclusion and shows hidden monotonic relationships.

Graph 7: Original vs Log-Transformed Distributions

Notebook cell 54



What it shows. Histograms of purchase_count, total_spend and avg_price per customer. Top row = original values, bottom row = after a log transform.

What we see in it.

- Original (top): almost all customers sit in the first bar and a very long right tail follows (purchase_count reaches 1,895; total_spend reaches about 57.7).
- After log (bottom): total_spend and avg_price become close to a bell shape. purchase_count is improved but still right-skewed with gaps, because it is a log of small whole numbers.
- Skewness drops from 5.24 to 0.37 (purchase_count), 6.28 to 0.09 (total_spend) and 4.20 to 0.05 (avg_price).
- Negative values on the log axes are normal: prices in this dataset are tiny scaled numbers (about 0.03), so their logarithm is negative.

Benefit. Justifies applying a log transform (and then scaling) before linear models, neural networks and clustering, and limits the influence of the extreme buyers. In the log scale, purchase_count and total_spend correlate at 0.96, which confirms the redundancy seen in Graphs 5 and 6.

2.1 Statistical checks printed in the notebook (no graph)

Skewness and kurtosis (cells 46 and 54)

Variable	Skew	Kurtosis	Skew after log	Kurtosis after log
age	0.61	-0.71	-	-
purchase_count	5.24	56.25	0.37	-0.65
total_spend	6.28	82.07	0.09	-0.50
avg_price	4.20	45.01	0.05	2.24

Benefit: numbers confirm what the histograms show. Age is only mildly skewed and needs no transform; the three behaviour variables are extremely skewed and do.

Outliers (cells 50, 51, 56)

- **Age:** range 16 to 99, mean 36.4, median 32; nothing below 16 or above 100.
- **Purchases per customer:** median 9, 90th percentile 60, 99th 187, 99.9th 385, maximum 1,895.
- **Top 10 buyers** made 1,143 to 1,895 purchases each, spread over 58 to 427 active days (up to 20.8 items per active day). *Reading:* this looks like genuine heavy buying, not data errors, so they are kept and handled with the log transform instead of being deleted.

- **Extra features tested (Spearman):** repeat_rate correlates 0.43 with active_days and 0.35 with items_per_active_day, so it adds partly new information; avg_price stays near 0 with all of them.

Benefit: separates real extremes from errors, so we do not remove valuable heavy customers by mistake.

Customer activity and repeat buying (cells 30 to 35)

Measure	Value
Unique customer-article pairs vs transactions	27,306,439 vs 31,788,324
Pairs bought more than once	3,515,413 (12.9% of pairs)
Distinct articles per customer	median 8, mean 20.0, max 1,346
Customers with only 1 article	154,947 (11.4%)
Customers with 5 or fewer / 10 or more	540,301 (39.7%) / 634,623 (46.6%)
Customers with 50+ / 100+ articles	145,352 (10.7%) / 40,537 (3.0%)
Product codes vs articles	47,224 product codes -> 105,542 articles (colour/size variants)
Products with 1 / 5+ / 10+ variants	24,066 / 3,914 / 963 (max 75 variants)

Benefit: the user-item matrix is very sparse and many customers have little history (cold-start risk), while a minority buy a lot. Repeat purchases exist, so 'buy again' candidates are worth including. Variants of the same product share one product_code, so items can be grouped.

Sampling and cleaning checks (cells 36 to 41, 57, 58)

- **10% customer sample** (seed 42): 137,198 customers, 3,162,621 transactions, 136,306 buying customers, 85,923 articles sold, 9,974 product images. All sampled transactions link to valid customers and articles (100%). Channel split is preserved (70.1% channel 2).
- **Consistency:** total_spend = purchase_count x avg_price (largest difference 7e-15); article_id // 1000 equals product_code for 100% of articles.
- **Duplicates:** 295,872 exact duplicate rows were counted in the sample (9.4%). The cleaning cell reports them but does not drop them; in retail data this can be the same item bought twice on one day, so it is worth a deliberate decision.

Benefit: proves the sample is faithful to the full data and the tables join correctly, so conclusions drawn on the sample can be trusted.

3. Part 2 and Part 3 EDA: coded but no saved output

None of the cells below contain saved results in the uploaded file, so no graphs or numbers are shown for them. The descriptions come from the notebook's own markdown and code. Part 2 uses the 10% cleaned sample.

Part 2: Extended EDA (cells 63 to 120)

ID	Graph	What it will show	Benefit
A1	Sales share by product group (bar)	Which broad groups dominate sales	Focus on groups that matter; baseline for popularity
A2	Top 15 product types (bar)	Concentration in a few types	Choose categories for features and for diversity checks
A3	Top 15 colours (bar)	Most sold colours	Colour is a useful item feature if concentrated
A4	Sales share by index group (pie)	Ladieswear, Menswear, Divided, Baby/Children, Sport split	Reveals segment imbalance
A5	Top 10 patterns (bar)	Whether plain items dominate	Decide if pattern adds signal
A6	Customer age histogram	Main age band and skew, with median line	Handle age properly (binning, imputation)
A7	Club status and news frequency (bars)	Whether one class dominates	May be low-information features
A8	Price raw and log10 (histograms)	Right-skew of price	Justifies a log scale for price
A9	Sales channel share (bar)	Store vs online split	Channel is a useful feature or filter
B1	Age group vs index group (heatmap)	Which age groups buy which segment	Shows age is useful for recommendations
B2	Price by product group (boxplot)	Price spread inside each group	Spot price outliers per group
B3	Channel mix per index group (stacked bar)	Segments with different channel habits	Channel-aware recommendations
B4	Median purchases by club status (bar)	Whether ACTIVE members buy more	Tests if membership predicts activity
B5	Median average price by age (line)	Price sensitivity across ages	Age-based price features
B6	Median purchases by age group (bar)	Most active age band	Target the most valuable groups
C1	Monthly transactions per index group (lines)	Trend and seasonality per segment	Segment-specific seasonal features
C2	Transactions by weekday (bar)	Weekend vs weekday behaviour	Day-of-week features
C3	Weekly transactions by year (lines)	Repeated seasonal peaks, 2019 vs 2020	Year-over-year seasonality and shifts
C4	Channel share per month (area)	Possible shift to online in 2020	Detects behaviour change over time
D1	Popularity long tail (cumulative curve)	Share of sales from top 100 / 1,000 / 5,000 / 10,000 articles	Shows if a popularity baseline can be strong
D2	Repeat rate by product type (bar)	Types people re-buy (socks, basics)	Justifies a 're-buy' recommender
D3	Cold start (printed numbers only)	Share of last-week buyers with no history; sales of unseen articles	Shows the need for non-personalised fallbacks
D4	Article lifespan (histogram)	Days between first and last sale	Short life means recent popularity wins
D5	Basket size (histogram)	Items per customer per day	Most baskets are small; shapes session design
D6	Days since last purchase (histogram)	Customer recency	Recently active customers are the likely buyers

Section E ('Key insights') in Part 2 is still an empty template in the notebook ('write in your own words after running').

Part 3: Churn definition and features (cells 124 to 140)

Churn is defined with a snapshot date T: an active customer bought in the 90 days before T; churn = 1 if there is no purchase in the 28 days after T. Two snapshots (train and test) keep the split time-aware.

ID	Graph	What it will show	Benefit
3.3	Churn share, train and test (bars)	Class balance of the target	Decides metrics (PR-AUC) and class weights
3.5a	Median of each feature by churn (table)	Which features differ between churners and others	Early look at predictive features
3.5b	Boxplots: days_since_last, n_last_28d, tenure_days	Separation of the two classes	Confirms the strongest predictors (recency, recent activity)
3.5c	Churn rate by age group (bar)	Whether age relates to churn	Decide if age is worth including
3.6a	Spearman heatmap of features and churn	Correlation among features and with target	Finds redundant features
3.6b	VIF (printed table)	Multicollinearity per feature (VIF above 10 = strong)	Which features to drop or combine
3.6c	Mutual information with churn (bar)	Non-linear predictive strength per feature	Ranks features beyond linear correlation
3.7	PCA scatter of PC1 vs PC2 coloured by churn	Structure in high-dimensional features and variance explained	Shows how separable churners are and how many components matter

Part 3 section 8 ('Key insights') is also an empty template. Part 4 (models) is outside EDA and is not covered here.

4. Key takeaways from the executed EDA

- **Time:** strong yearly seasonality plus isolated one-day spikes. Use a time-based split and recent popularity; ignore the partial first and last months.
- **Missing data:** structural for FN and Active (fill with 0); tiny for age, club status, news frequency and product description. Nothing needs to be dropped.
- **Skew:** purchase_count, total_spend and avg_price are heavily right-skewed; log-transform them before scaling and modelling.
- **Redundancy:** purchase_count, unique_articles, total_spend and active_days move together (0.80 to 0.99); keep one or combine. avg_price is independent and worth keeping.
- **Customers:** many have very little history and a minority buys a lot; heavy buyers look genuine and should be kept. About 13% of customer-article pairs are repeat purchases.
- **Catalogue:** a few attributes dominate (black, solid, upper body) and rare categories should be grouped into 'Other'.